

AI Literacy Statement

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1 Conceptual Understanding

Generative AI is a broad category which refers to a class of machine learning models which produce unique outcomes from a set of training data. They are built on neural networks, which take a set of inputs and make predictions based off of patterns recognition techniques. At each node in the network in the prediction is assigned a weight which helps the model determine what the most likely outcome is. The most common generative AI tools consumers come across are Large Language Models (LLMs). LLMs take a text input and use contextual information to determine what answer makes the most sense.

In essence, LLMs, such as ChatGPT, Claude, Gemini, and others, are prediction machines, taking the patterns in our language and using them to respond to their users in ways that appear logical. These tools have many benefits. They can be incredibly useful for simple or repetitive tasks such as translation or quick data analysis, which may result in significant error if conducted by a human. However, there are still limitations to these technologies. Taking translation, for example, an LLM often will have difficulty understanding the nuance of a language in the same way as a native speaker, leading to error that may change the meaning of a phrase. Additionally, concerns about hallucinations have arisen, where LLMs may present false information in a manner which appears factual. For people who may be using LLMs to learn about a topic with which they are not familiar or to strengthen their

knowledge in their area of study, they may be privy to accepting this false information as true. Creatives also have reservations about the use of LLMs due to potential for plagiarism. Because training data often includes copyrighted material, responses produced by LLMs may not be wholly original and may rely heavily on these sources to produce predicted outcomes. Hence, while AI has the potential to offer great benefit, there are important concerns which need to be addressed.

2 Practical Skills

Using AI to improve efficiency requires a strong understanding of how these tools are beneficial and where they fall short. Throughout my study, I've learned tools and strategies to help counteract some of the limitations of our current models, integrating their use into my work without relinquishing my ownership of it.

One of the greatest concerns I have about using LLMs is hallucination. The error rates in responses, even from the best publicly-available models is high enough to raise eyebrows when doing in-depth research or analysis. To mitigate this, I employ retrieval-augmented generation (RAG) tools, such as Google's NotebookLM. RAG limits the amount of source material that is available to a model. This decreases hallucination because the parameters force the model to take information from user-input sources. This method also allows for easier fact checking, as claims made by the model can be directly verified in the uploaded sources.

Another method I use to ensure effective use of LLMs is Chain-of-Thought (CoT) prompting. This method is a simple way to improve the reasoning of one's model, by including in the prompt some variation of the phrase "think through this step-by-step." In including this phrase, the model will be more transparent about the inferences it makes to reach its final conclusion and is more likely to catch logic errors. CoT prompting also works against the metaphorical black-box of AI, by forcing the LLM to demonstrate more of the steps it uses

to reach a conclusion.

What these two strategies have in common is the implementation of checks to the AI ensuring that it works as expected. These methods allow me to target my AI usage and make sure that it doesn't overstep or provide me with inaccuracies. I don't use AI for extensive writing because I've found it to be largely ineffective at returning large blocks of text, but it can be valuable in the knowledge acquisition stage of research. It's greatest strengths are in pattern recognition, so using it to correct grammar, track argument development, or parse data tend to be more valuable uses.

3 Ethics Statement

As AI continues to permeate our workflows, it's important to make sure that we retain cognitive sovereignty. Helen Edwards, co-founder of the Artificiality Institute, defines this as "maintain[ing] intellectual agency while working symbiotically with AI" (Edwards, 2025). This point touches on one of the biggest ethical problems we face currently: separation of human work from AI. Taking a Luddite's stance on AI is futile, especially with its already widespread use. Thus, implementation methods need to be generated to ensure safe and effective use of these powerful tools.

Some of the greatest ethical concerns around AI, regarding privacy, originality, and training bias, can be attributed in part to a lack of regulation. Having a technology, such as this, unregulated leads to valid concerns about its safety and quality. Legislation needs to be implemented to ensure that there is oversight over the development and implementation of these models, protect people who may not want their data used in training, and standardize blame assignment in cases where AI causes unexpectedly negative outcomes. Google DeepSeek developer, Mustafa Suleyman, emphasized recently that "we really are at a turning point in the history of humanity" and he's correct (Suleyman & Bhaskar, 2023). It's how we evolve from this moment which will define the current era. The argument could even be

made that greater regulation will foster growth and innovation in unique ways (Mariyono & Hidayatullah, 2025).

I don't think AI is going to replace us. Science fiction has toyed with the idea of technological takeover for a long time, largely drawing the conclusion that human connection is difficult to replicate. In 1909, E.M. Foster wrote a short story called "The Machine Stops," which depicts a world in which everything has become mechanized and humans have largely lost their social nature. When one of the characters starts to question why this is, he proclaims "man is the measure," meaning that as much as technology may influence humanity, each one of us is still in control of how and where it does so (Foster, 1928). The social structures we currently have are built on trust, which aren't easily replaced with AI. Thus, we will continue to grow alongside technology, but ultimately will decide what role it plays in our shared future.

References

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